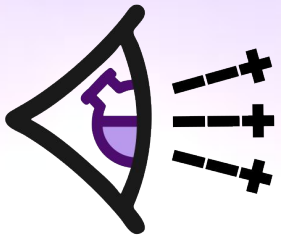
SEND-SLAM: Democratizing the access to Simultaneous Location And Mapping algorithms

Independent Research Project



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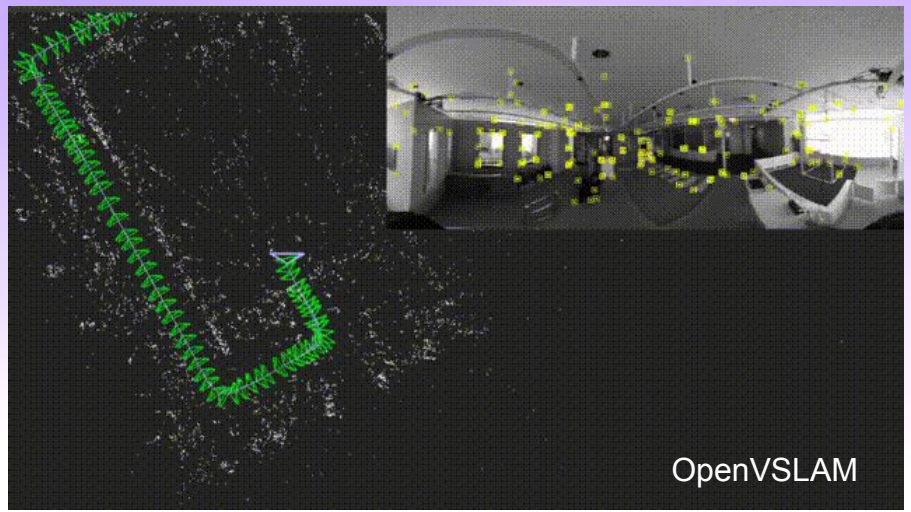


What is the SLAM problem?

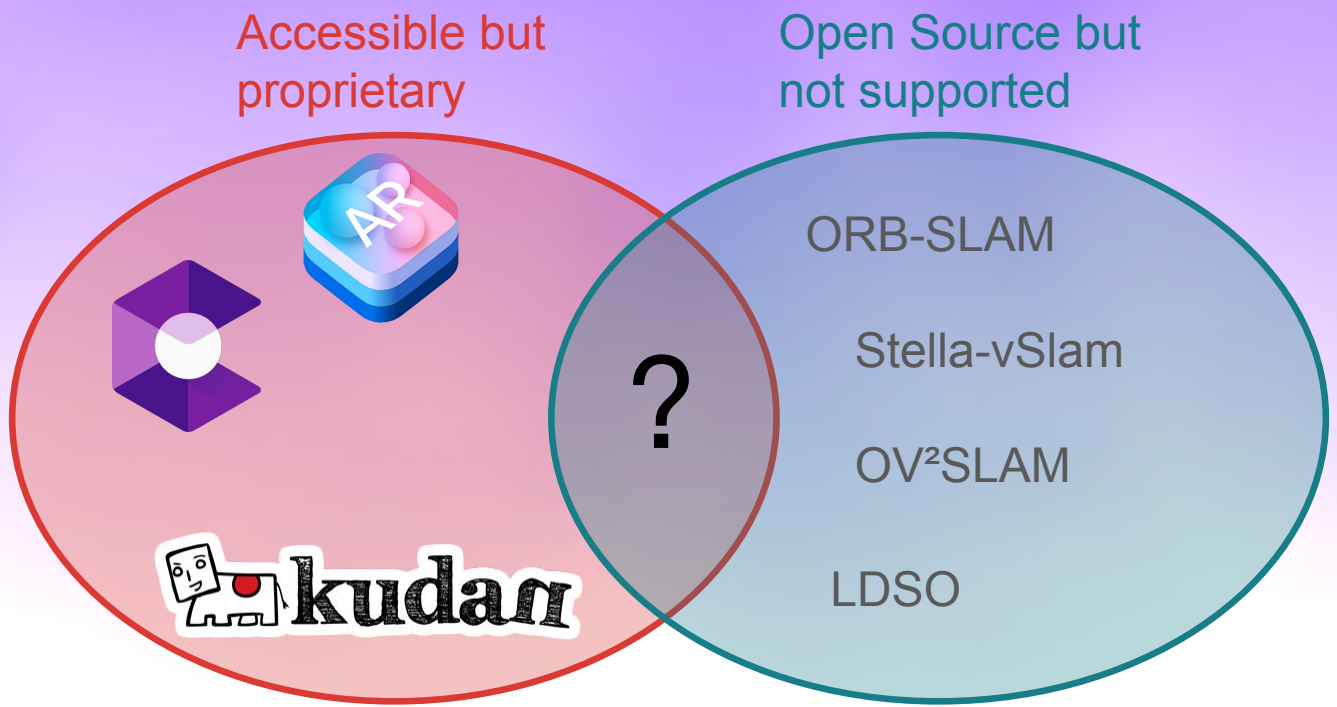
Simultaneous Location And Mapping:

Given a series sensor observations over discrete time steps, the SLAM problem is to compute an estimate of the agent's state and a map of the environment.

- SLAM can be done with lots of sensor combinations.
- We focused especially on cameras, commonly called vSLAM



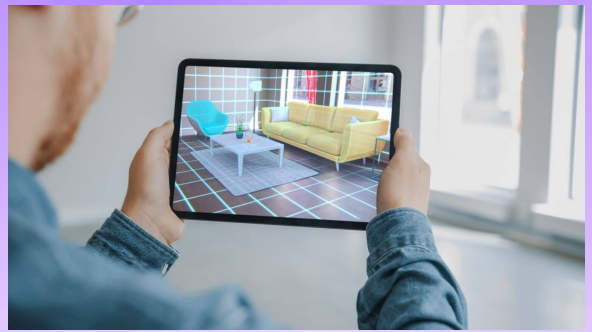
State of SLAM algorithms



SLAM can be found everywhere!



XR Headsets



Phones and Tablets



Drones



Self-Driving Cars



Objectives Of Research

- Can we make these already great algorithms available for everyone?
- Can we create a sort of SLAM framework that is more developer friendly?

Yes!*

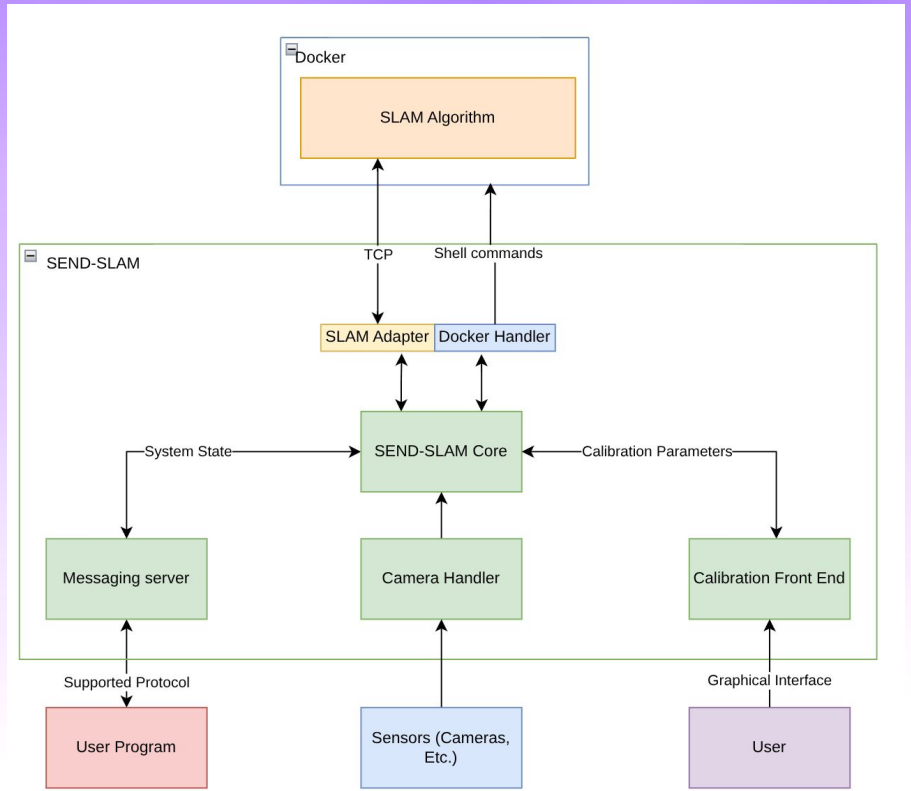
*Although it’s still a work in progress.



Our solution:

- We found an interesting paradigm, **Message Passing.**
- I wanted to learn how to program in Elixir.
- We sought to solve the dependency hell that is needed to run SLAM algorithms in a computer.
- We created a developer friendly interface for people to develop their programs on top.

The architecture



Limitations/Future Research

- The system cannot reliably process more than 30fps for a camera.
- Expand on the number of cameras and other sensor configurations.
- Improve on the developer facing side to expose more useful data to the client programs.
- Implement different SLAM algorithms to be used with the system
- Using Docker introduces a slight overhead and another dependency

Thanks! Any questions? / Enjoy a Live demo

https://github.com/SoloFausto/SEND-SLAM

